■ Customer Churn Analysis

Telecom Customer Retention | Exploratory Data Analysis + Machine Learning

Dataset	Model	Accuracy	Churn Rate
7,032 customers	Logistic Regression	~79-81%	~26-27%

■ Project Objectives

Identify	Which customers are most likely to churn
Analyze	The root causes and behavioral patterns behind churn
Predict	Build an ML model to forecast churn with high accuracy
Recommend	Actionable business strategies to reduce customer loss

■ Dataset Overview

The dataset is the publicly available **IBM Telco Customer Churn** dataset containing information about 7,032 unique telecom customers across demographics, services, account details, and billing.

Feature	Type	Description
customerID	ID	Unique customer identifier
tenure	Numeric	Months the customer has been with the company
Contract	Categorical	Month-to-month / One year / Two year
MonthlyCharges	Numeric	Current monthly charge amount
TotalCharges	Numeric	Total amount billed to date
InternetService	Categorical	DSL / Fiber optic / No
PaymentMethod	Categorical	Electronic check / Mailed check / Auto
Churn	Target	Yes / No — whether the customer left

■ Data Preprocessing

Before analysis, the data was cleaned using the following steps:

```
df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')

# Convert TotalCharges from string to numeric
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Remove rows with missing values
df.dropna(inplace=True)
```

✔ Converted **TotalCharges** from text/string to numeric — some entries had blank spaces causing parse errors.

✔ Dropped **11 rows** with NaN values after conversion (negligible data loss).

✔ Result: **7,032 clean customer records** ready for analysis.

■ Exploratory Data Analysis (EDA)

1. Customer Churn Distribution

```
sns.countplot(x='Churn', data=df)

plt.title('Customer Churn Distribution')
```

Figure 1: Churn Distribution — ~73% stayed, ~27% churned

■ Key Insights

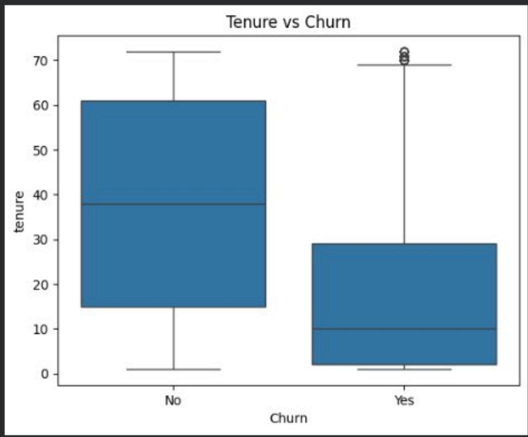
- Approximately **26–27% of customers churned** — roughly 1 in 4.
- This class imbalance is important to consider when training the ML model.
- Businesses must act on this segment to prevent revenue loss.

2. Tenure vs Churn

```
sns.boxplot(x='Churn', y='tenure', data=df)

plt.title('Tenure vs Churn')
```

```python
sns.boxplot(x='Churn', y='tenure', data=df)
plt.title("Tenure vs Churn")
plt.show()
```



In our graph: For Churn = Yes → median ≈ 10 months For Churn = No → median ≈ 38 months

💡 Meaning:

👉 Half of churned customers stayed: Less than 10 months 👉 Half stayed: More than 10 months

"The boxplot shows that customers who churn have significantly lower tenure, indicating that new users are more likely to leave early."

Figure 2: Tenure Distribution for Churned vs Retained Customers

■ Key Insights

- Churned customers have a **median tenure of ~10 months** vs ~38 months for retained customers.
- **New customers are at the highest churn risk** — the first year is critical.
- Long-tenured customers are far more stable and loyal.

3. Monthly Charges vs Churn

```
sns.boxplot(x='Churn', y='MonthlyCharges', data=df)

plt.title('Monthly Charges vs Churn')
```

Figure 3: Monthly Charges for Churned vs Retained Customers

Key Insights

- Churned customers have a **higher median monthly charge (~$79)** vs retained (~$61).
- Premium/higher-tier users are more likely to leave — possibly due to unmet expectations.
- Value-for-money perception is a critical churn driver.

4. Contract Type vs Churn

```
sns.countplot(x='Contract', hue='Churn', data=df)

plt.title('Contract Type vs Churn')
```

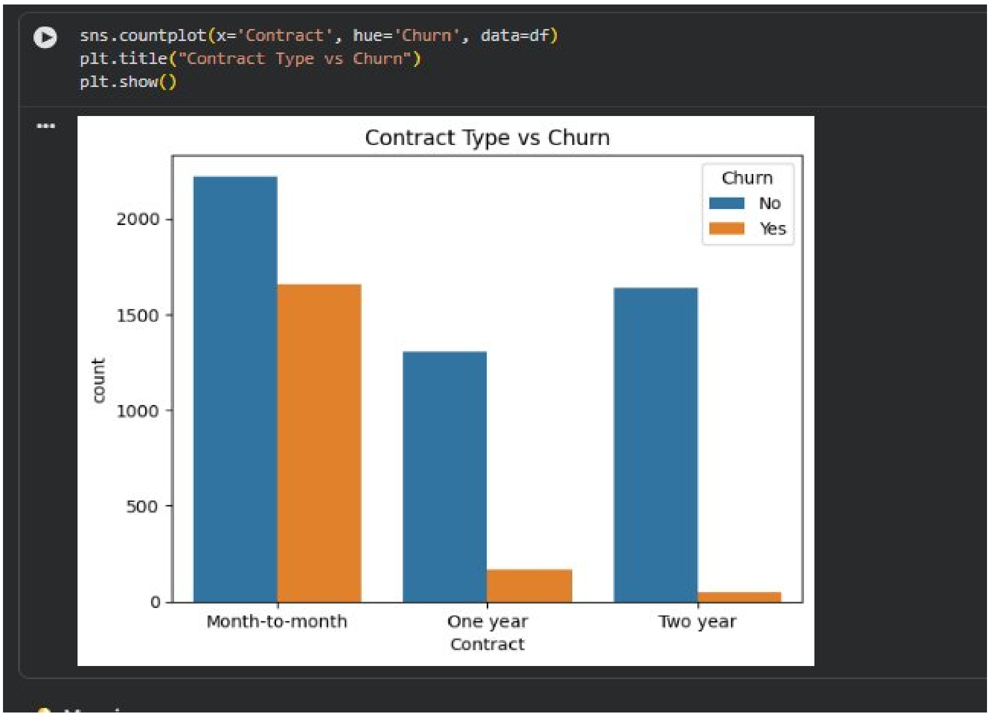


Figure 4: Churn Rate Breakdown by Contract Type

■ **Key Insights**

- **Month-to-month customers have the highest churn** — nearly 43% of them leave.
- One-year contract customers have moderate churn (~11%).
- **Two-year contract customers almost never churn** (~3%) — locking in customers works.

■ Machine Learning Model

A **Logistic Regression** model was trained to classify customers as likely to churn or not, using all available features after encoding.

Step 1: Encode Categorical Variables

```
df_ml = df.copy()
df_ml.drop('customerID', axis=1, inplace=True)
df_ml = pd.get_dummies(df_ml, drop_first=True)
# Converts categorical columns into binary (0/1) numeric columns
```

Step 2: Train / Test Split + Model Training

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

X = df_ml.drop('Churn_Yes', axis=1)
y = df_ml['Churn_Yes']

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

accuracy = model.score(X_test, y_test)
print(f"Model Accuracy: {accuracy:.2%}") # ~79-81%
```

Model Accuracy	Train Size	Test Size	Algorithm
~79-81%	80% (5,625 rows)	20% (1,407 rows)	Logistic Regression

Step 3: Feature Importance

```
importance = pd.Series(model.coef_[0], index=X.columns)
importance.sort_values().tail(10).plot(kind='barh')
plt.title("Top Factors Influencing Churn")
```

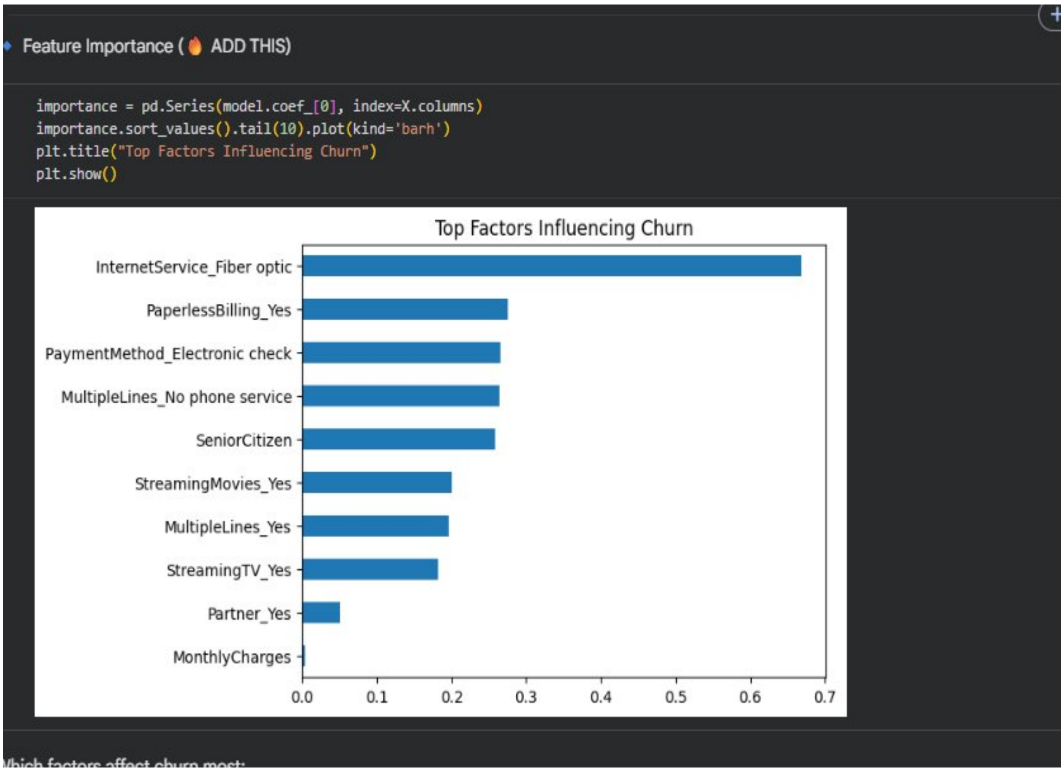


Figure 5: Top 10 Features Influencing Churn (Logistic Regression Coefficients)

■ **Key Insights**

- **Fiber optic internet service** is the single strongest predictor of churn.
- **Paperless billing** and **electronic check payment** are associated with higher churn.
- **Senior citizens** and customers with **multiple lines** also churn more.
- **Having a partner** is a mild protective factor against churn.

Step 4: Predicting New Customer Churn

```
new_customer = pd.DataFrame({
'tenure': [5],
'MonthlyCharges': [90],
# ... all other feature columns
})

new_customer_encoded = pd.get_dummies(new_customer)
new_customer_encoded = new_customer_encoded.reindex(
```

```
    columns=X.columns, fill_value=0)

prediction = model.predict(new_customer_encoded)
probability = model.predict_proba(new_customer_encoded)

print(f"Will Churn: {prediction[0]}")
print(f"Churn Risk: {probability[0][1]:.2%}")
```

■ Outputs a binary **Yes/No** churn prediction for any new customer.

■ Also returns a **churn probability score** — enabling risk tiering.

■ Key Insights & Business Recommendations

■ Summary of Key Insights

Factor	Finding	Risk Level
Customer Tenure	Median churn tenure = 10 months; retained = 38 months	■ High
Monthly Charges	Churners pay ~$18 more per month on average	■ Medium
Contract Type	Month-to-month churn ~43%; Two-year churn ~3%	■ High
Internet Service	Fiber optic users have highest churn risk	■ High
Payment Method	Electronic check users churn more than auto-pay	■ Medium
Senior Citizens	Higher churn rate compared to non-seniors	■ Medium
Partner / Dependents	Customers with partners/dependents churn less	■ Protective

■ Business Recommendations

■	**Improve Onboarding**	Launch personalized welcome programs for new customers in the first 3–6 months to build habit and loyalty.
■	**Promote Long-Term Plans**	Offer targeted discounts (10–20%) to month-to-month customers to switch to annual/bi-annual contracts.
■	**Enhance Fiber Quality**	Invest in fiber optic service reliability and speed — address the service issues driving churn in this segment.
■	**Proactive Outreach**	Use the churn model to identify high-risk customers monthly and send personalized retention offers.
■	**Loyalty Rewards**	Reward long-tenured customers with exclusive discounts, priority support, or free feature upgrades.
■	**Promote Auto-Pay**	Encourage customers on electronic checks to switch to auto-pay — it correlates with lower churn rates.

■ Business Impact

■ Proactively identify high-risk customers before they leave

■ Run targeted retention campaigns with measurable ROI

■ Reduce revenue loss — retaining a customer costs 5x less than acquiring a new one

■ Improve customer satisfaction through personalized engagement

■ Conclusion

This project successfully analyzed **7,032 telecom customer records** to understand and predict customer churn. Using exploratory data analysis, we identified that **tenure, monthly charges, contract type, and internet service type** are the strongest predictors of churn. A Logistic Regression model trained on this data achieves **~79–81% accuracy**, enabling businesses to proactively identify at-risk customers and take data-driven retention actions.

By combining machine learning with clear business insights, this approach has the potential to **significantly reduce churn rates**, improve customer lifetime value, and drive sustainable revenue growth.